5. **In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

A colony of bees is being studied.

The number of bees in the colony at the start of the study was 30 000

Three years after the start of the study, the number of bees in the colony is 34 000

A model predicts that the number of bees in the colony will increase by $p\%$ each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.

Assuming the model,

(a) find the value of $p$, giving your answer to 2 decimal places.

**(3)**

According to the model, at the end of $N$ years of study the number of bees in the colony exceeds 75 000

(b) Find, showing all steps in your working, the smallest integer value of $N$.

**(5)**